

DIGITAL TWIN CATALOG

API and Frontend Connector

Read-only REST API, tile to database link, Cesium viewer and Postman collection.

Project	Digital Twin Catalog (NARAGA 3D Catalog), reusable 3DCityDB template
Document	API and Connector
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Introduction

Status: **draft v0.1 for review** · Base path `/api/v1` · Read-only · JSON · OpenAPI at `/api/v1/openapi.json`,
UI at `/api/v1/docs` (django-ninja).

Endpoints

Endpoint	Purpose
GET /health	Service + database status
GET /catalog? lod=&theme=&dataset=&bbox=minx,miny,maxx,maxy&status=published&limit=&offset=	Layers with active tileset (source: catalog.v_layer)
GET /layers/{id}/tileset	Active tileset of a layer
GET /datasets , GET /datasets/{code}	Datasets and their layers per LOD
GET /datasets/{code}/stats	Object counts per class/LOD
GET /features/{objectid}	Attributes of one city object from citydb (name, class, height, function, usage, generic attributes, LODs available, dataset, lineage)
GET /features?dataset=&bbox=&q=&limit=	GeoJSON footprints + key attributes for search

Rules: `bbox` is lon/lat EPSG:4326; `lod` 0–4, empty = all; `status` default `published` ; pagination `limit` (default 50, max 500) + `offset` ; unknown filter value → 422.

Error format (all errors):

```
{ "error": { "code": "invalid_parameter", "message": "lod must be between 0 and 4", "details": {"field": "lod"} } }
```

Catalog item:

```
{ "layerId": "uuid", "dataset": {"code": "oikn", "name": "OIKN", "generatedBy": "external", "attribution": null}, "theme": "building", "lod": 1, "status": "published", "featureCount": 0, "isStale": false, "bbox": [116.6, -1.0, 116.9, -0.8], "tileset": {"version": 1, "provider": "cesium_ion", "ionAssetId": 123456, "url": null, "heightOffsetM": 0} }
```

(values illustrative)

Tile ↔ database link

Each tile object must carry its `objectId` (gml:id). The frontend reads it on click and calls `/features/{objectId}`. Verified status: **pending** for real tiles (the API side is tested with a fixture feature) (convertwin id preservation is checked on 9 Oct).

Connector deliverables

- `client/naraga-catalog.js` - ES module, no dependencies: `listLayers({lod})`, `loadTileset(viewer, lod)`, `getFeature(id)`, `enablePicking(viewer)`.
- `viewer/index.html` - simple Cesium viewer served at `/viewer/`: API URL + ion token inputs (token only in browser `localStorage`), LOD/theme/dataset selectors, layer metadata (status, version, stale), click → attributes, error banner.
- `postman/` - `naraga-catalog.postman_collection.json`, `local` and `railway` environments (`baseUrl`, `secret ionToken`). Tests per request: status code, schema, filter respected, pagination, error format, response-time budget. Run headless: `newman run postman/naraga-catalog.postman_collection.json -e postman/local.postman_environment.json`.

Cesium ion token lives only in the frontend. NARAGA token auth is a later phase.

Verification status

Verified on 4 October 2026 against the local stack: 13 pytest integration tests (filters, pagination, error format, tileset, features, admin audit user) and the Postman collection through newman (21 requests, 61 assertions, 0 failed). The viewer loads, lists layers from the API and shows a clear message for a layer without an active tileset. Loading real tiles and click-to-attributes is not yet tested because no tileset exists yet.